# Operation Midnight Hammer: an STK Model

Claudio Vestini

December 13, 2025

## 1 Introduction

This document reports the modelling of the United States Department of War mission codenamed *Operation Midnight Hammer*, which involved aerial strikes on several Iranian nuclear facilities. The operation was modelled using the Ansys software package Systems Tool Kit (STK) [1]. The work was carried out using `STK 12.10` locally on a Razer Blade 14 (Windows 11) machine. The plots were generated using the `Report & Graph Manager` toolbox included in the STK package.

## 2 Mission goals & objectives

Operation Midnight Hammer (OMH) was a United States-led military strike on Iranian nuclear facilities, executed by the United States Air Force and Navy under the command of United States Central Command. The overall objective of OMH was to "severely degrade Iran's nuclear weapon infrastructure", according to General John Daniel Caine, chairman of the Joint Chiefs of Staff. The operation was carried out on the 21st and 22nd June 2025 over 37 hours, with most strikes occurring between 02:10 and 02:30 Iran Standard Time on June 22nd (18:40-19:00 EDT on June 21st), as part of the broader Iran-Israel war that began with Israeli strikes on 13 June 2025. The strikes were authorised by President Donald Trump following failed diplomatic efforts, including a proposed 60-day negotiation period announced in February 2025, and amid International Atomic Energy Agency concerns over Iran's uranium enrichment to 60 per cent purity (not far from weapons grade). This marked the first United States offensive action in this conflict and the first attack on Iranian territory since 1988.

## 3 Mission concept overview

The mission involved 13 B-2 Spirit stealth bombers supported by 112 US aircraft, one US nuclear submarine, and a total of 75 surface-to-surface Tomahawk missiles, 14 GBU-57A/B Massive Ordnance Penetrator (MOP) bombs, and several air-to-surface weapons [2]. A central component contributing to the success of the mission was space intelligence, which involved the use of the SBRIS constellation for real-time missile launch detection and cueing, the GPS constellation for navigation and tracking of aircraft and weapons, and several electro-optical LEO satellites for post-strike battle damage assessment (BDA).

The mission unfolded as follows: at 04:00:00 UTC (00:00 EDT) of January 21st 2025, two packages of B-2 bombers took off from the 509th Bomb Wing at Whiteman Air Force Base, MO. The first package, acting as a decoy, comprised 6 aircraft and headed west to Oklahoma, California, Hawaii, and finally Guam. Its sole objective was to deceive enemy intelligence. The real operational package, comprising 7 B-2 units (each manned by two pilots), headed east on an 18-hour trajectory towards Iran. It traversed the Atlantic Ocean and the Mediterranean Sea, performing a total of 3 mid-air refuellings over international waters enabled by 12 KC-135 Stratotankers. Upon entering Israeli airspace, the seven B-2 aircraft were joined by a support package comprising F-35s, F-22s, and other aircraft to aid the strike operations by neutralising potential Iranian countermeasures (no firings were reported). At roughly 21:00:00 UTC (17:00 EDT), a US submarine supported by Israeli warships stationed in the Persian Gulf fired a total of 75 sub-orbital Tomahawk missiles directed at the Isfahan Nuclear Technology Centre, coordinating the strike with the bombers and attack package. With the Tomahawks still soaring through the sky, at roughly 22:00:00 UTC, the joint flight formation entered Iranian airspace through the Iraqi border. At 22:40:00 UTC (18:40 EDT) on June 21st 2025, the lead B-2 bomber, codenamed Mytee-11, released

the first two GBU-57s on the Fordow facility, closely followed by another 10 MOPs plus an additional 2 released on Natanz (2 per aircraft). Simultaneously, all GBU-57 warheads and Tomahawk missiles hit their intended targets, in a window of roughly 20 minutes. By 23:30:00 UTC (19::30 EDT), the B-2 bombers had left Iranian airspace, completely undetected. The aircraft returned to Whiteman AFB from their direction of origin, aided by multiple aerial refuellings, arriving at around 18:30:00 UTC (the total flight time was 37 hours). The mission is summarised in Figure 3.1.

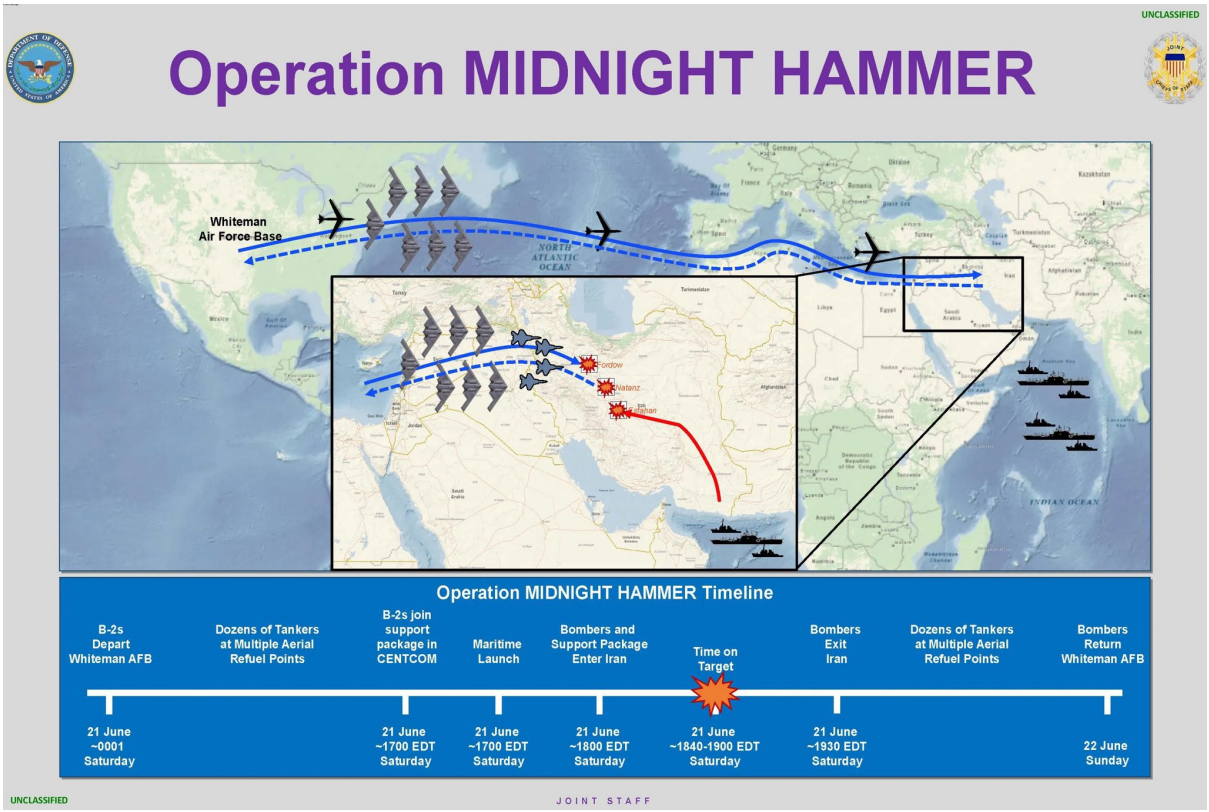



Figure 3.1: Official Unclassified mission plan for Operation Midnight Hammer (from the DOW).

# 4 STK model

The STK model of OMH presented in this report is based on publicly available information, including declassified DOW documents [2], a report by the National Security Space Association (NSSA) [3], interviews of army personnel directly and indirectly involved in the operation, and several news outlets reports and reconstructions. Publicly unavailable details were approximated using data from analogous missions, and, throughout the report, we explicitly state when approximations were made. Our model is comprised of a total of 65 individual STK objects, and covers the entire mission timeline through an STK scenario starting on the 21st June 2025 at 04:00:00.000 UTC and ending on the 22nd Jun 2025 at 19:00:00.000 UTC. The model includes the 7 B-2 Spirit live bombers, the 6 decoy B-2 bombers, a total of 6 KC-135 Stratotankers for mid-air refueling, the GBU-57 bombs and Tomahawk missiles, the USS Georgia nuclear submarine and ISN Rahav Israeli ship, several facilities for launch and communications, and a total of 32 satellites and sensors for intelligence and guidance. All figures, source LaTeXcode, source data files and source STK code is made available at the supporting repository:

ClouD-161803/MAE341-FinalProject.

## 4.1 Aircraft and refuelling operations

A simplified overview of the mission, which hides all components except for the 7 live bombers, all bombs and missiles, the take-off and landing Whiteman Air Force Base in Missouri, the naval infrastructure, and the targets at Fordow, Natanz, and Isfahan, is shown in Figure 4.1. This simplified view shows the strike aircraft trajectory (dark blue), which was reconstructed using a total of 534 individual waypoints

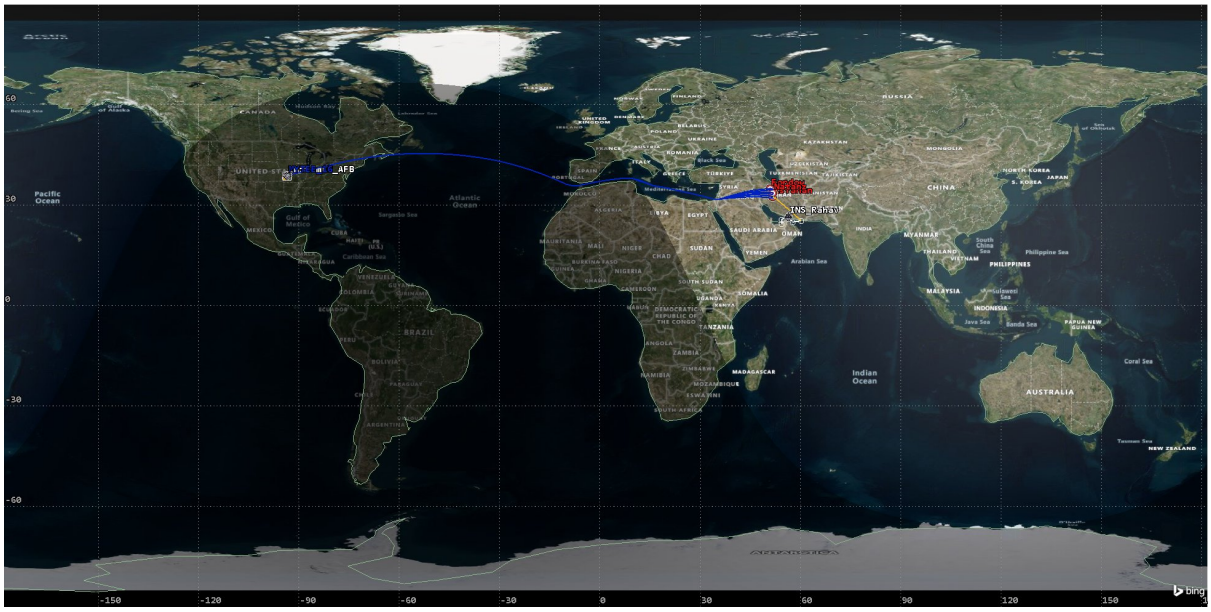



Figure 4.1: Simplified mission overview for Operation Midnight Hammer in STK.v



Figure 4.2 (a) 3D CAD model of the B-2 Spirit aircraft. Taken from [4].




Figure 4.2 (b) STK zoomed view.

based on the report at [2]. The B-2 Spirit bombers were modelled using STK's built-in `Aviator` propagator module, which enables the planning of multi-phase aircraft missions involving missile and bomb launches, take-off and landing manoeuvres, and mid-air refuelling operations. The 3D models of the B-2 planes, unavailable natively on STK, were pulled from the FetchCFD website at [4] and modified using CAD in Solidworks (using an Oxford University student license) to be made compatible with the STK environment. The 3D model adopted, as well as its appearance in STK, are shown in Figure 4.2a and Figure 4.2b, respectively. The bombers were assumed to cruise at $0.25\,\mathrm{km\,s^{-1}}$ (or 500 knots, which corresponds to Mach 0.85), the B2's optimal operational speed (the speed which minimises fuel usage), at a (reported) altitude of 12 km.

During the mission, a total of 6 refuelling events occurred (3 refuellings per leg). According to [2], the first refuelling operation took place over US soil, the second over the Atlantic Ocean (in international waters), and the third over the Mediterranean Sea (also in international waters), with the precise locations still classified. Information about where the tanker aircraft took off, and their trajectories, is also classified. It was assumed that the outbound leg refuellings occurred over Ohio, the $-30°$ longitude meridian on the great circle trajectory over the Atlantic, and in the waters south of Crete, respectively, and that the return leg refuellings occurred over the same locations, but in reverse. These approximations were informed by unclassified B-2 refuelling locations for Operation Iraqi Freedom (2003) and by the proximity of these locations to US Air Force and Navy bases. In particular, we modelled the tanker aircraft as

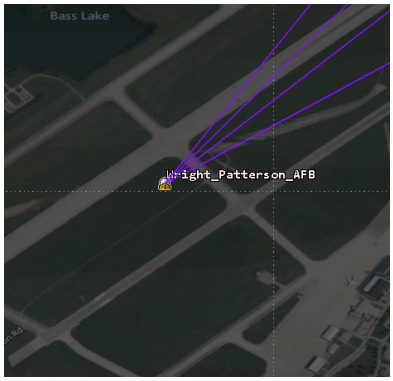



Figure 4.3 (a) W. Patterson AFB.




Figure 4.3 (b) Lajes Field, Portugal.




Figure 4.3 (c) Incirlik AFB, Türkiye.

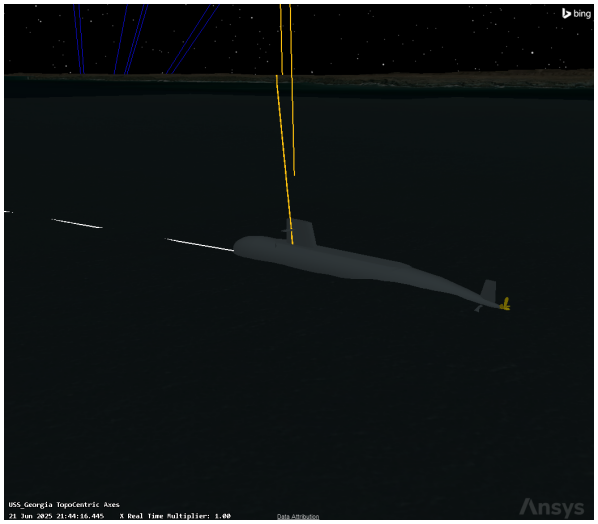



Figure 4.4 (a) Closeup of the USS Georgia model.

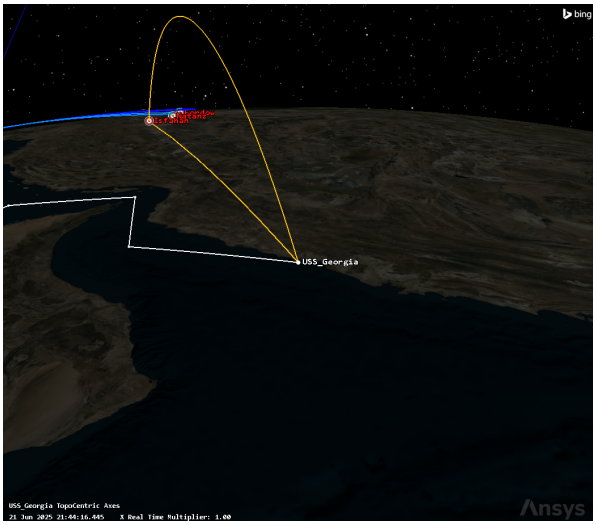



Figure 4.4 (b) Tomahawks missiles launched on Isfahan.

KC-135 Stratotankers taking off from Wright-Patterson Air Force Base, OH; Lajes Field, located off the coast of Portugal; and Incirlik Air Base in Türkiye, respectively. The facilities created over these bases are shown in Figure 4.3b, Figure 4.3b, and Figure 4.3b, respectively. The precise coordinates of these air force bases were retrieved from the Air Force website at [5], and the refuelling operation locations were approximated as those of the Iraqi Freedom mission. The refuelling was modelled using `Aviator`, with the loiter orbit chosen as a small figure-8 orbit for each aircraft and a speed of $0.05\,\mathrm{km\,s^{-1}}$. The decoy package of 6 B-2 Spirit bombers was modelled using aircraft STK objects propagated using `GreatCircle`. Their trajectory was made to follow the reported flight route outlined in Section 3. The support package's takeoff, rendezvous and landing locations, as well as its exact composition, are all classified. Since these aircraft played no relevant active role and had no reported impact on the mission's success, we did not include them in our STK model.

## 4.2 Maritime forces and Tomahawk missiles

The attack on Isfahan was carried out remotely by a joint fleet of US and Israeli ships and submarines. The precise identity of the assets used is classified, but both the USS Georgia nuclear submarine and the ISN Rahav Israeli warship were reported to have taken part in the operation. Thus, these are the two maritime forces present in our model. Although the trajectory of these vessels was not disclosed in any of the official reports, we assumed they were jointly deployed in the Persian Gulf and modelled them as ship objects (propagated using `GreatArc`) in STK. Both vessels reached their target destination off the southern coast of Iran, opening fire at 21:00:00 UTC. A closeup of the USS Georgia model, as well as the trajectory of the Tomahawk missiles it fired on Isfahan (orange trajectory), are shown in Figure 4.4a and Figure 4.4b, respectively. In this figure, the ISN Rahav ship model is hidden for visibility. The Tomahawk missiles were modelled using the STK missile object, with a ballistic trajectory launching from the warships onto the Isfahan Nuclear Technology Centre and a fixed delta-v of $3.235\,\mathrm{km\,s^{-1}}$.




Figure 4.5: GBU-57 MOP bomb being transported. The bomb measures 6 m in length. ABC News.

## 4.3 GBU-57 bunker buster bombs

The attacks on Fordow and Natanz were carried out by the main B-2 Spirit package discussed in Section 4.1. The bombs dropped on these two targets were GBU-57 MOP (also known as "bunker busters"), 13 608 kg warheads which can penetrate up to 61 m of reinforced concrete. These bombs are only produced in the United States, and three-quarters of the total US stock was deployed in a single night during OMH (a total of 14 bombs were dropped). One such bomb is shown in Figure 4.5 with human beings for size comparison. We modelled these bombs as air-dropped missile objects in STK, with zero delta-v (since the bombs were dropped rather than launched), and a drop location 12 km ahead of their targets (this was calculated using the ballistic trajectory of the GBU-57 bombs, which have a terminal velocity of 317 $\mathrm{m\,s^{-1}}$, and the B-2's cruising speed of 250 $\mathrm{m\,s^{-1}}$ and initial altitude of 12 km. Using these values, the GBU-57 was estimated to stay in the air for approximately 48 s covering 12 km downrange). The trajectories of the 14 bombs released over Fordow and Natanz are shown in Figure 4.6 (green). In total, 6 of the 7 B-2s released their warheads over Fordow (their trajectories are shown in dark blue in the figure), while the remaining bomber released its two bombs over Natanz (light blue). All times for release and manoeuvres were modelled to exactly match the real mission, discussed in detail in Section 3.

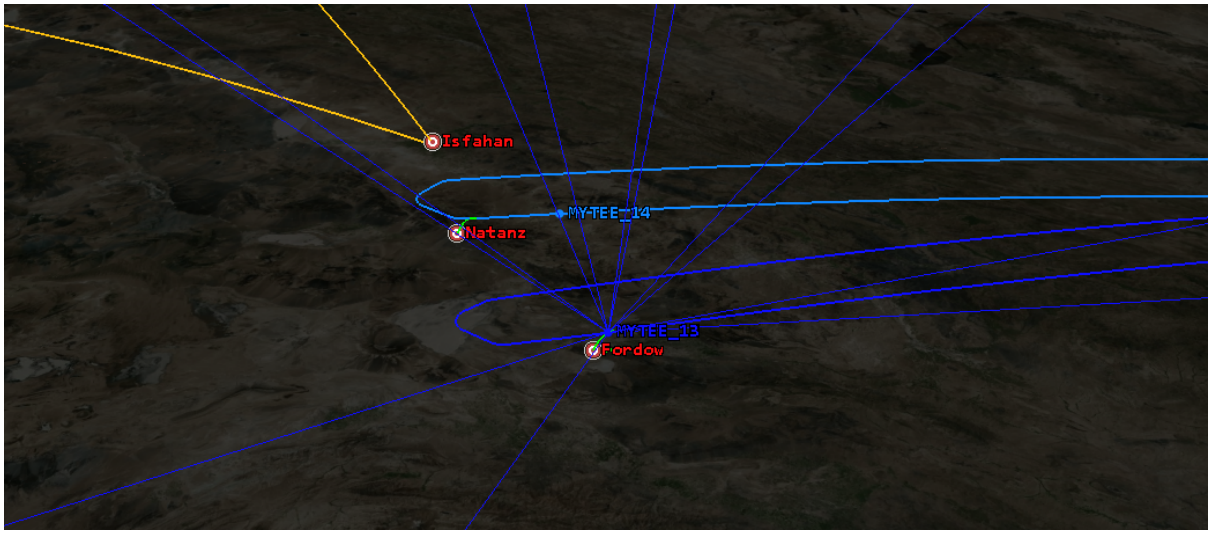



Figure 4.6: Aerial strike trajectories of GBU-57 MOP bombs dropped from B-2 spirit bombers.

## 4.4 Satellites and intelligence

As discussed in Section 1, satellite intelligence represented the central coordinating element in this mission. We employed a total of 32 satellites to enable lightspeed communication among the several aircraft involved, to track and guide all bombs and missiles, and to support BDA imaging of the sites post-strike.

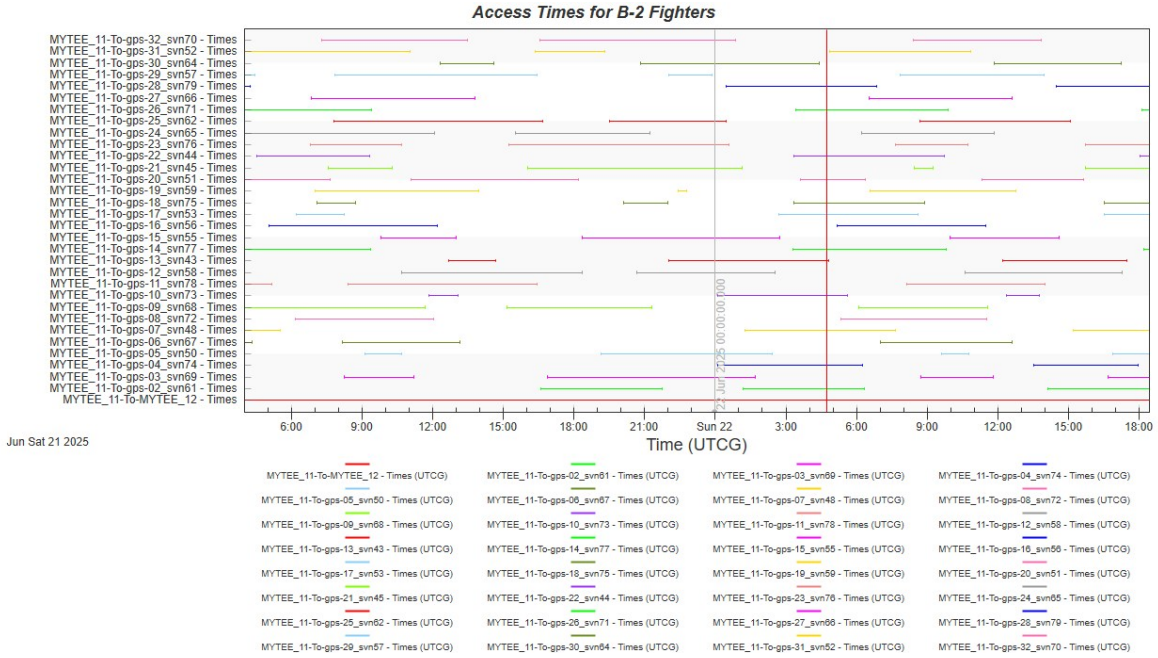
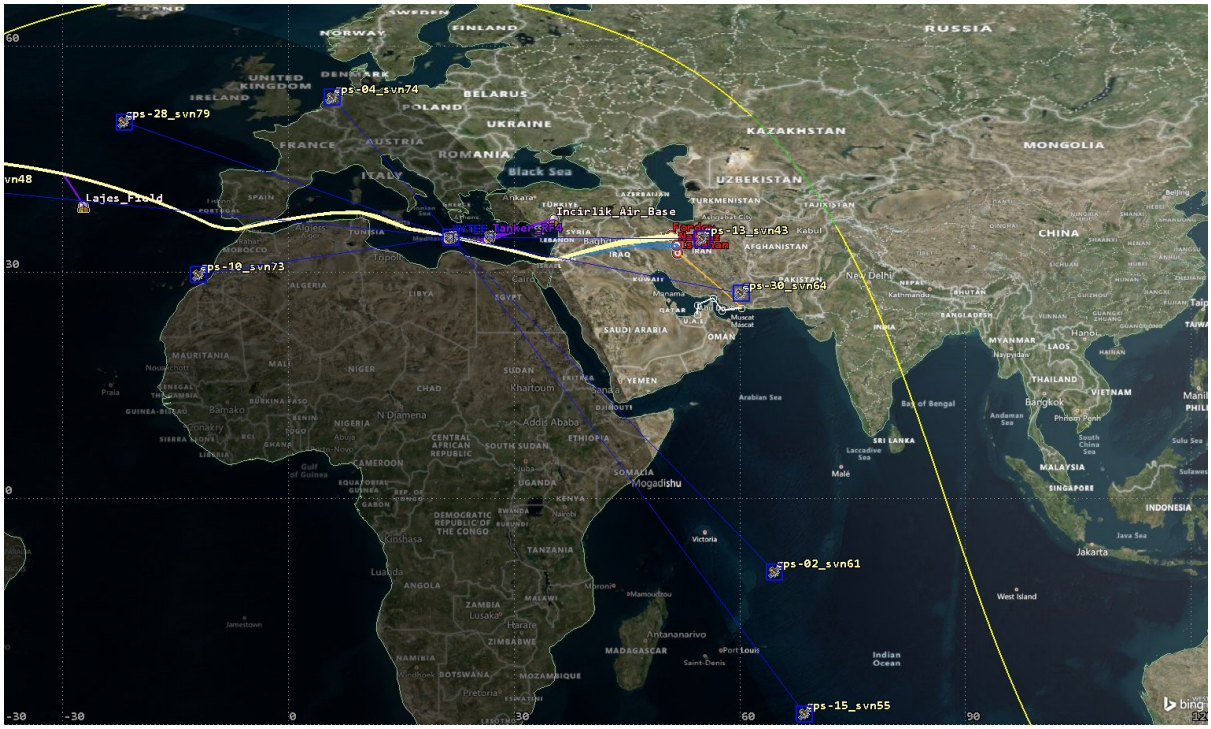



Figure 4.7: Access times for the lead B-2 spirit bomber (codename MYTEE-11) to the GPS constellation.

To coordinate the missile strikes between the B-2 bombers and the USS Georgia and the ISN Rahav maritime forces, we employed the SBRIS constellation in STK, which granted continuous access between the two forces throughout all mission-critical phases. The route of the B-2 bombers was also continuously tracked using the GPS constellation: access times between the lead bomber and GPS satellites are shown to give continuous coverage in Figure 4.7. This enables the onboard positioning and guidance system to provide continuous corrections to the route of the B2 bombers, enabling them to follow the predetermined mission path. In addition, GPS positioning is crucial for the coordination of refuelling maneuvers, which demand the KC-135 Stratotankers to approach the B2 bombers with incredibly accurate matching of both position and velocity, to accurately dock the fuel pipe into the stealth aircraft's conduit. The communication access prior to the third outbound refuelling maneuver is displayed in Figure 4.8 to showcase the centrality of space intelligence throughout this mission phase.

Further, the GPS constellation was used to guide the GBU-57 bombs and the Tomahawk missiles. These warheads have very narrow mission durations, and their access times are shown in Figure 4.9a and Figure 4.9b, respectively. Although brief, these communication windows were essential to the success of the mission: the Tomahawk missiles have an active control system which guided all 75 units to their intended target on the surface of Isfahan. Similarly, the GBU-57 bombs have a passive, GPS-enabled grid-fin control system used to correct the course of the warheads to their exact impact location. BDA imaging (discussed in greater detail later) confirmed that, out of 14 released MOPs, all 14 hit their predetermined coordinates with an error below 5 m. This unforseen degree of precision is entirely enabled by space intelligence, and is unprecedented in military history.

For BDA imaging, the documentation available as of the writing of this report does not include any details about the exact satellites used in OMH. However, there is no doubt that some low-earth orbit (LEO) electro-optical satellite was employed to image the strike locations both before and after impact (we later show the satellite images reported by several news outlets). As a result, we approximated this mission phase, employing a LEO orbit satellite with custom orbital parameters selected to provide two overhead passes before and after the strikes in the daylight (altitude 500 km, inclination 67.7053°, RAAN 224° and other parameters zero at the starting epoch). These passes are shown in Figure 4.10 and Figure 4.11, respectively. The satellite was also equipped with a downward-facing sensor with conical


gps-28_svn79
gps-04_svn74
gps-10_svn73
gps-13_svn43
gps-30_svn64
gps-02_svn61
gps-15_svn55
Lajes_Field
Incirlik_Air_Base
RAF_Fairford
Diego_Garcia
RUSSIA
KAZAKHSTAN
MONGOLIA
CHINA
UKRAINE
BELARUS
ROMANIA
Black Sea
UZBEKISTAN
TURKMENISTAN
AFGHANISTAN
IRAQ
SYRIA
SAUDI ARABIA
OMAN
YEMEN
Arabian Sea
Bay of Bengal
SRI LANKA
INDONESIA
MALAYSIA
SINGAPORE
THAILAND
VIETNAM
Jakarta
Bangkok
Mogadishu
ETHIOPIA
SUDAN
Khartoum
Addis Ababa
TANZANIA
Antananarivo
MADAGASCAR
Indian Ocean
New Delhi
Kabul
Baghdad
Cairo
Ankara
Tripoli
Algiers
Rabat
Nouakchott
Bamako
Abuja
Kinshasa
Luanda
Lusaka
Harare
Pretoria
Windhoek
Jamestown
Port Louis
Victoria
Nairobi
Kampala
Kigali
Maputo
Gaborone
Muscat
Manama
Doha
Male
Laccadive Sea
Andaman Sea
South China Sea
Java Sea
Sulu Sea
Celebes Sea
West Island
Saint-Denis
Mamoudzou
Maroni
Hanoi
Phnom Penh
Naypyidaw
Kathmandu
Thimphu
Dhaka
Yangon
Vientiane
Laos
Myanmar
Bhutan
Nepal
Bangladesh
Sri Lanka
Malaysia
Singapore
Indonesia
Vietnam
Thailand
Philippines
Taiwan
Beijing
Tianjin
Shanghai
Chongqing
Guangzhou
Shenzhen
Hong Kong
Macau
Kunming
Guiyang
Nanning
Hainan
Sichuan
Yunnan
Guizhou
Hunan
Hubei
Henan
Shaanxi
Gansu
Qinghai
Xinjiang
Tibet
Inner Mongolia
Heilongjiang
Jilin
Liaoning
Hebei
Shanxi
Shandong
Jiangsu
Anhui
Zhejiang
Fujian
Jiangxi
Guangdong
Guangxi
Hainan
Taiwan
Philippines
Sulawesi
Java Sea
Sulu Sea
Celebes Sea
South China Sea
Andaman Sea
Laccadive Sea
Arabian Sea
Bay of Bengal
Indian Ocean
Black Sea
Caspian Sea
Red Sea
Persian Gulf
Gulf of Oman
Gulf of Aden
Gulf of Guinea
Mediterranean Sea
Aegean Sea
Adriatic Sea
Tyrrhenian Sea
Ionian Sea
Balearic Sea
Ligurian Sea
Gulf of Lion
Bay of Biscay
Irish Sea
North Sea
Baltic Sea
Norwegian Sea
Barents Sea
Greenland Sea
Iceland
Ireland
United Kingdom
France
Spain
Portugal
Italy
Germany
Poland
Sweden
Finland
Norway
Denmark
Belgium
Netherlands
Switzerland
Austria
Hungary
Czechia
Slovakia
Slovenia
Croatia
Bosnia and Herzegovina
Serbia
Montenegro
Albania
North Macedonia
Greece
Turkey
Cyprus
Lebanon
Israel
Jordan
Syria
Iraq
Iran
Kuwait
Saudi Arabia
Qatar
United Arab Emirates
Oman
Yemen
Afghanistan
Pakistan
India
Nepal
Bhutan
Bangladesh
Myanmar
Thailand
Laos
Vietnam
Cambodia
Malaysia
Singapore
Indonesia
Philippines
Brunei
Taiwan
China
Mongolia
Russia
Kazakhstan
Uzbekistan
Turkmenistan
Tajikistan
Kyrgyzstan
Afghanistan
Pakistan
India
Nepal
Bhutan
Bangladesh
Myanmar
Sri Lanka
Maldives
Malaysia
Singapore
Indonesia
Philippines
Brunei
Taiwan
China
Mongolia
Russia
Kazakhstan
Uzbekistan
Turkmenistan
Tajikistan
Kyrgyzstan

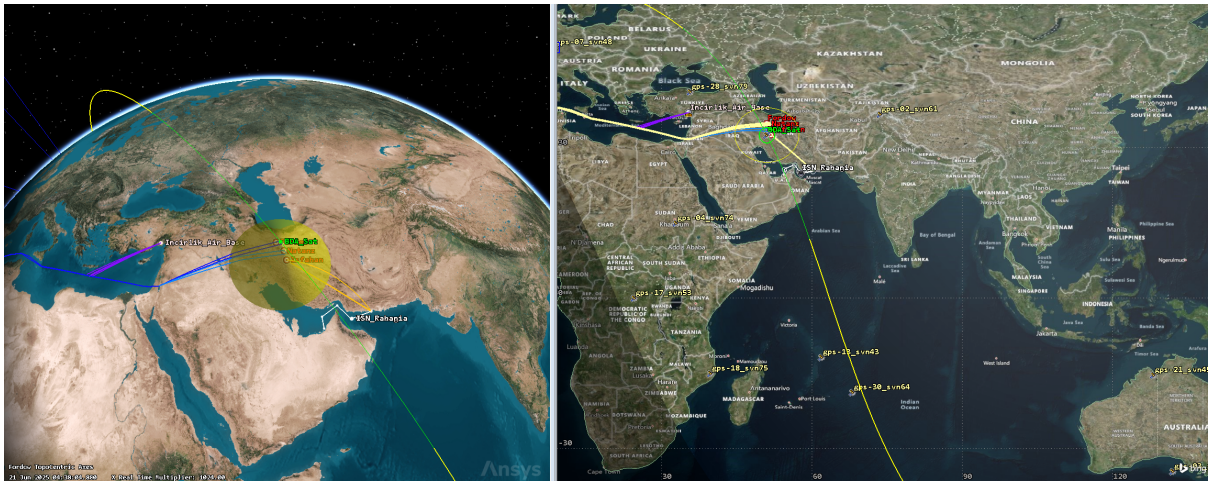


Figure 4.10: Pre-strike imaging by the LEO electro-optical satellite.

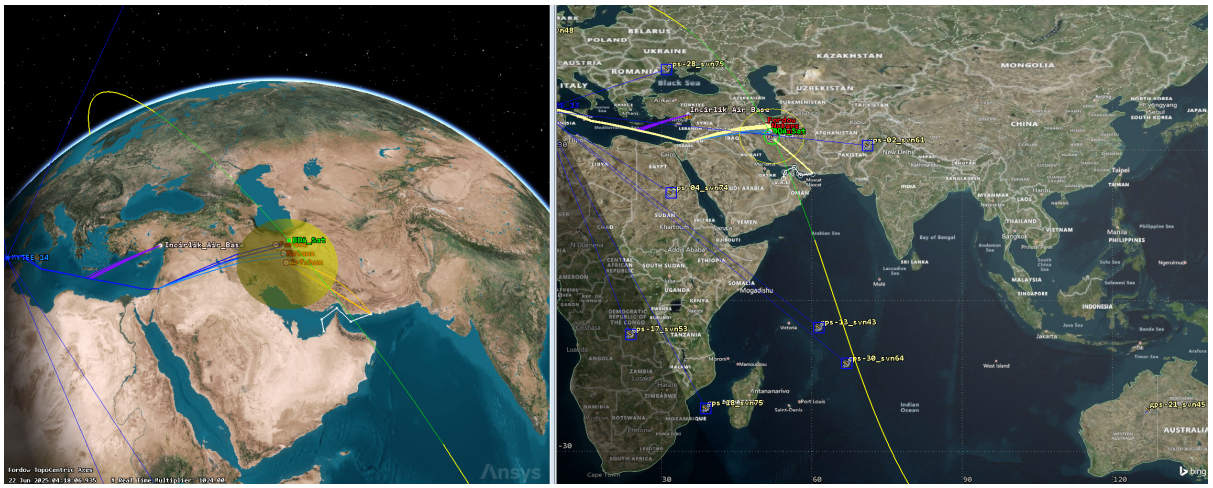


Figure 4.11: Post-strike imaging by the LEO electro-optical satellite.

our model to the real-world mission.

# 5 Conclusion

This document presented a model of Operation Midnight Hammer in STK. Apart from classified or otherwise inaccessible information, to the best of the authors' knowledge, this model reproduced precisely the unfolding of the operation, following each critical mission phase with minute accuracy. The model highlighted several of the mission's tradeoffs: first, we noted that, although the US has capable supersonic aircraft, the mission was carried out by subsonic B-2 bombers to prioritise stealth over mission duration. The mission lasted 37 hours in total, which was demanding for the 14 pilots involved, but it also ensured that US forces maintained the element of surprise. Secondly, reliance on extensive aerial refuelling, which occurred six times with tankers from Wright-Patterson Air Force Base in Ohio, Lajes Field in the Azores, and Incirlik Air Base in Türkiye, introduced operational complexity and heightened the risk of detection. However, this approach enabled global reach without the need for forward basing, thereby mitigating geopolitical tensions. The project also highlighted the meticulous planning required to execute such a complex mission, and the central role that satellite intelligence plays in coordinating so many aircraft, ships, missiles and ground facilities. Coordinating aerial strikes with maritime missile launches over several thousand kilometres, with no direct line of sight due to Earth's curvature, is only possible with the aid of lightspeed satellite communications. These mission details often go unnoticed in press coverage of military operations, despite their fundamental role.

Throughout this project, I developed a profound appreciation for the coordination and operational

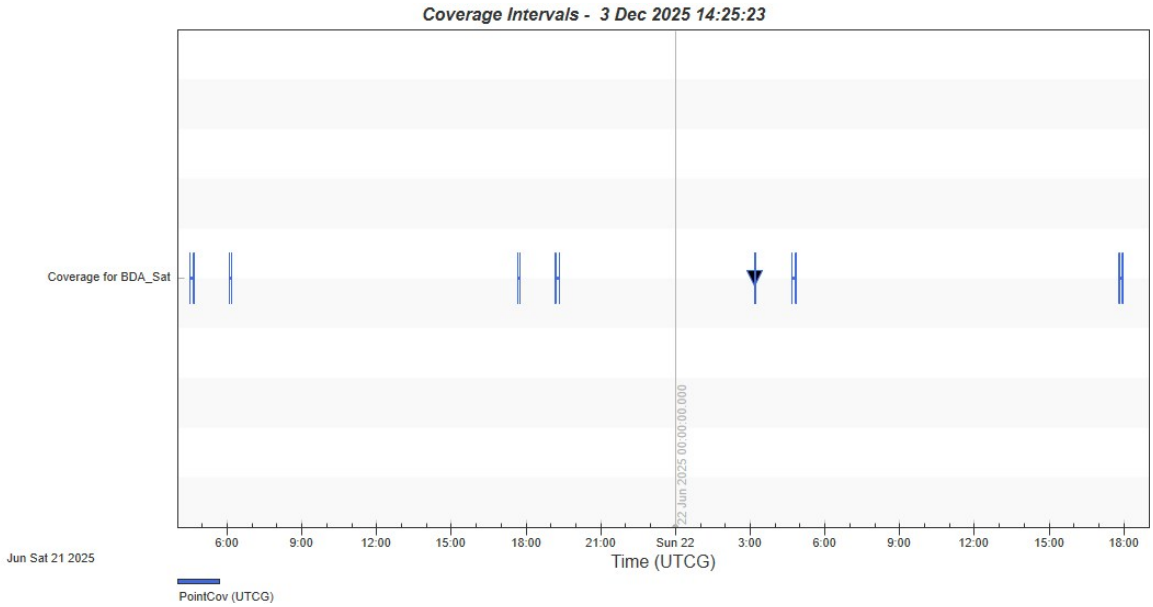



Figure 4.12: STK coverage graph for the LEO electro-optical satellite over the three facilities.

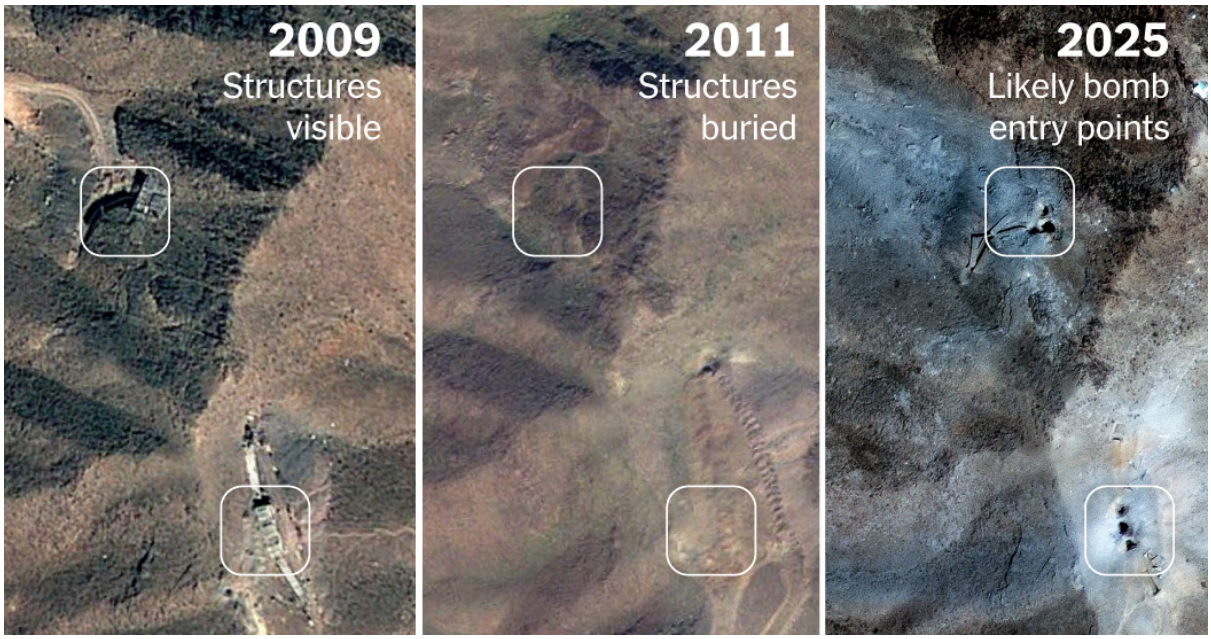



Figure 4.13: Post-strike BDA images over Fordow. Credit: New York Times.

capabilities of the US Air Force, and a deep understanding of how these missions are facilitated through the use of satellites. As an enthusiast on military operations and assets—especially those of the united states—I have never been exposed to the level of intricacy required in planning a mission as complicated as OMH. Even just by simulating the mission in STK, without any stakes or human lives on the line, turned out to be an incredibly stimulating challenge in careful planning and meticulous mission timing.

# References


[1] Ansys, Inc. *Systems Tool Kit (STK)*. Version 12.10. Canonsburg, PA, 2025. URL: https://www.ansys.com/products/missions/ansys-stk.

[2] LeMay Center for Doctrine Development and Education. *Doctrine Paragon: Operation MIDNIGHT HAMMER*. Tech. rep. Accessed: 2025-12-02. Curtis E. LeMay Center for Doctrine Development and Education, July 2025. URL: https://www.doctrine.af.mil/Portals/61/Jul-25-Doctrine%20Paragon-Operation%20Midnight%20Hammer.pdf.

[3] Chris Williams and Sean Kirkpatrick. *National Security Space Contributions to Operation Midnight Hammer*. Tech. rep. Accessed: 2025-12-03. National Security Space Association, Moorman Center for Space Studies, June 2025. URL: https://nssaspace.org/wp-content/uploads/2025/06/midnight.hammer.pdf.

[4] Max Payne. *B-2 Spirit 3D Model in 3D*. https://fetchcfd.com/threeDViewGltf-embed-project/4932-b-2-spirit-3d-model. Original 3D model by Muhamad Mirza Arrafi. Licensed under CC Attribution. Accessed: 2025-12-03. n.d.

[5] Steve Ingram. *Base Map*. https://www.af.mil/News/Art/igphoto/2000790881/. Map created by Steve Ingram of the Air Force News Agency. Accessed: 2025-12-03. n.d.